The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Supplementary Material

Appendices and supplementary figures accompanying the article. Section, equation and figure numbers referenced here follow the main text; supplementary figures are numbered S1, S2, ...

A Notation and units

These neural-mass models are phenomenological: potentials are in mV, firing rates in s^{-1} (≡Hz), time constants in ms, and frequencies in Hz, unless noted; a "–" marks a dimensionless quantity. A few glyphs are reused across the four models (JR, LaNMM, NMM2/MPR, QIF) by long-standing convention; these are disambiguated by context and flagged below. In particular, A without an argument is the excitatory PSP *gain* in the JR/LaNMM synapses; $A(t)$ is the instantaneous envelope of the two-tone field and A_Ω a lock-in amplitude. The applied-field modulation amplitude is ε in JR and LaNMM alike, and A_f in NMM2/QIF. These are not interchangeable units. In JR and LaNMM the field is a membrane polarization in mV entering the sigmoid argument, so ε is measured against the sigmoid voltage scale $1/\rho \approx 1.8$ mV, which is the same in both models. In NMM2/QIF the field enters the excitatory membrane current in the dimensionless MPR units of v_E, and A_f carries no volt scale at all. Likewise C is the JR connectivity constant and the NMM2 global synaptic gain $C\,(\equiv J)$.

Symbol	**Meaning**	**Units**
Firing-rate nonlinearity (sigmoid)		
$\sigma(v)$	population firing-rate transfer function	s^{-1}
$\sigma'(v)$	sigmoid slope (read-out gain)	$s^{-1}\,mV^{-1}$
$\sigma''(v)$	sigmoid curvature; the memoryless detector coefficient	$s^{-1}\,mV^{-2}$
$F(Q)$, F', F''	MPR steady-state rate curve, Eq. (5), and its derivatives	–
$C(Q,\omega)$, $K(\omega)$	MPR second-order DC coefficient at fixed current, Eq. (13), and at fixed voltage excursion, Eq. (15); d, ν the population's relaxation rate and ringing frequency	–
e_0	half the maximal firing rate ($\sigma \to 2e_0$ as $v \to \infty$)	s^{-1}
v_0	sigmoid inflection (half-maximal) potential	mV
ρ	sigmoid slope parameter	mV^{-1}
v^*	operating point (field-free fixed point of v)	mV
$v = y_1 - y_2$	net pyramidal membrane potential (LFP proxy)	mV
PEIX	Population Excitation–Inhibition Index (a σ'-based brain-state index)	–
Single-column Jansen–Rit		
A, B	excitatory, inhibitory PSP gains	mV
a, b	excitatory, inhibitory synaptic rates (inverse time constants)	s^{-1}
y_1, y_2	excitatory, inhibitory post-synaptic potentials	mV
C, $C_1 - C_4$	intracolumn connectivity constants	–
p	external input rate; JR bifurcation parameter	s^{-1}
Applied TI field and coupling		
ε	field modulation amplitude, JR and LaNMM (a membrane polarization entering the sigmoid argument)	mV
A_f, $I_E(t)$	amplitude of the field's current-like forcing $I_E(t) = P\,E(t)$ in the E voltage equation, NMM2/QIF (dimensionless MPR units of v_E)	–
m	modulation depth	–
f_c, $\omega_c = 2\pi f_c$	carrier frequency	Hz, $rad\,s^{-1}$
f_1, f_2	the two applied carrier tones ($f_c = (f_1 + f_2)/2$)	Hz
Δf, $\Omega = 2\pi\Delta f$	envelope (beat) frequency, $\Delta f = f_2 - f_1$	Hz, $rad\,s^{-1}$
$E(t)$, $s(t)$	applied field; membrane perturbation reaching the nonlinearity ($s = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}$, Eq. (8))	$V\,m^{-1}$; mV
λ, λ_0^c, λ_f^c	field-to-membrane coupling: generic, DC (door c), and carrier-effective ($\lambda_f^c = \lambda_0^c \lvert H_c \rvert$)	mV / ($V\,m^{-1}$)
$\Lambda[\hat{E}]$	direct (non-synaptic) field term in the sigmoid argument, Eq. (3); equals $s(t)$ in the JR column	mV
$H_c(\omega)$, H_{gate}	access filter of compartment c; optional channel-kinetic filter	–
$H_s(\omega)$	JR synaptic transfer function, §2.5	mV s

Symbol	Meaning	Units
$H_m(\omega),\ \tau_m$	effective membrane low-pass and its time constant	–; ms
$\tau_s,\ \tau_{\mathrm{rel}}$	synaptic and population relaxation times in the adiabatic reduction	ms
s_{syn}	filtered firing rate in Eq. (6), distinct from field polarization $s(t)$	s^{-1}
τ	effective access time used in the filter comparisons	ms
Network dynamics, response, and bifurcation		
$\lambda_\pm = -\gamma \pm i\omega_0$	leading Jacobian eigenvalue pair	s^{-1}
γ	damping of the critical mode ($\propto$ distance to the Hopf)	s^{-1}
$\omega_0,\ f_0 = \omega_0/2\pi$	natural (Hopf) frequency	$\mathrm{rad\,s}^{-1}$, Hz
$\tau_{\mathrm{rev}} = 1/\gamma$	network reverberation time	s
$\mathcal{Q} = \omega_0/2\gamma$	near-Hopf resonance quality factor	–
$G(\Omega)$	response to a generated beat drive; $\propto 1/\gamma$ near resonance	input/output dependent
D	beat-source coefficient for the injection convention in Eq. (21)	model-dependent
A_Ω	demodulated response: lock-in amplitude at Ω of v (mV) or, in §4.7 and the QIF network, of a rate	mV or s^{-1}
Y_Ω, Y_{LF}	detector (pre-network) output at Ω and at baseband, §2.4	–
$\mathcal{M}[\cdot]$	peak-to-peak excursion of a periodic quantity, Eq. (18)	–
η_{det}	TI/tACS drive ratio of a memoryless transfer at matched peak polarization, App. D	–
δ	threshold offset of a step detector, App. D	mV
$\ell(\omega)$, ℓ_0	AC space constant and its DC value, Eq. (23)	μm
PPC	pairwise phase consistency of single-neuron spike phases at Δf [1]	–
A_{open}	same lock-in, open loop (detector only, no recurrence); $A_\Omega/A_{\mathrm{open}}$ dimensionless	mV
I, Q	in-phase, quadrature lock-in components	mV
C^*, J^*	coupling at the Hopf (criticality)	–
Laminar model (LaNMM)		
P_1, P_2	deep (alpha) and superficial (gamma) pyramidal populations	–
PV, SS, SST	parvalbumin, stellate, somatostatin interneurons	–
Exact mean field (NMM2 / MPR) and QIF network		
r_E, r_I	population firing rates (E, I)	s^{-1}
v_E, v_I	mean membrane potentials (E, I), in dimensionless MPR units	–
$\eta_j,\ \bar{\eta}$	per-neuron excitability; mean excitability (= common drive, bifurcation param.)	–
Δ	Lorentzian half-width of η_j (excitability heterogeneity)	–
$\tau_E, \tau_I, \tau_a, \tau_g$	E, I membrane and AMPA, GABA synaptic time constants	ms
$C\ (\equiv J)$	global synaptic coupling (inter-population gain)	–
$A_{ee}, A_{ei}, A_{ie}, A_{ii}$	synaptic weight matrix $(1, 1, 1, 2)$	–
s_E, s_I	second-order synaptic activations; $\hat{K}_a, \hat{K}_g$ their operators	s^{-1}
$V_j,\ V_{\mathrm{peak}}$	QIF neuron potential; spike/reset threshold	mV
N	neurons per population (QIF; $N = 8000$)	–
$A_\Omega/\langle r_E \rangle$	Δf modulation depth (envelope-locked timing metric)	–

Table 1: Symbols, meanings, and (nominal) units. Reused glyphs are disambiguated by context; see the head note on A and C.

B Simulated models: equations and parameters

The models are written in the NMM1 transfer-function formalism of Eq. (3) [2]: synaptic operators $\hat{L}$ transduce firing rates into PSPs, the PSPs sum with any direct field term $\Lambda[\hat{E}]$ in the sigmoid argument, and the sigmoid returns the rate. The Jansen–Rit column is given here; the laminar model is specified in [3, 4] with its field drive in Eq. (26), and the NMM2 PING motif and its QIF network in Eq. (27) and §3.4.

Jansen–Rit column

Writing y_0 for the excitatory PSP driving the interneurons, y_1 and y_2 for the excitatory and inhibitory PSPs onto the pyramidal population, external input p, and the applied-field perturbation $s(t)$ (§2.2) entering the pyramidal sigmoid,

$$\ddot{y}_0 = Aa\,\sigma\big(y_1 - y_2 + s(t)\big) - 2a\dot{y}_0 - a^2 y_0, \tag{1}$$

$$\ddot{y}_1 = Aa\,[\,p + C_2\,\sigma(C_1 y_0)\,] - 2a\dot{y}_1 - a^2 y_1, \tag{2}$$

$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{3}$$

integrated as six first-order ODEs; the LFP proxy is $v = y_1 - y_2$ [5].

Table 2: Jansen–Rit parameters [5].

Symbol	Meaning	Value
A, B	excit./inhib. PSP gain	3.25, 22 mV
a, b	synaptic rates	100, 50 s^{-1}
v_0, e_0, ρ	sigmoid inflection, max rate/2, slope	6 mV, 2.5 s^{-1}, 0.56 mV^{-1}
C	global connectivity	135
C_1, C_2, C_3, C_4	couplings	C, $0.8C$, $0.25C$, $0.25C$
p	external input (bifurcation knob)	0–400 Hz

C The cubic Hopf normal form lacks a quadratic mixer

The Stuart–Landau equation [6]

$$\dot{z} = (\alpha + i\omega_0)z - |z|^2 z + \mathcal{F}(t), \qquad \gamma = -\alpha > 0, \tag{4}$$

describes a complex mode z with damping γ and natural angular frequency ω_0, driven by the applied input $\mathcal{F}$. For a weak input $\mathcal{F} = \varepsilon f(t)$ on the stable side, the perturbative response has the form $z = \varepsilon z_1 + \varepsilon^3 z_3 + \cdots$. There is no second-order source: the first-order response contains only the applied frequencies, and the cubic first contributes at order ε^3. Thus the normal form has a resonant linear susceptibility but lacks the quadratic difference-frequency generation of Eq. (10). This statement concerns the weak-input expansion; it is not an exclusion of every higher-order intermodulation or strong-drive response.

In the numerical control ($\alpha = -0.5$, $f_0 = 10$ Hz, $f_c = 100$ Hz, $\varepsilon = 0.5$), a square-law detector gives a beat-frequency lock-in amplitude of 2.5×10^{-1}, while $\mathrm{Re}\, z$ gives approximately 2×10^{-8}, at the numerical floor. Driving the same mode directly in band recovers the forced $1/\gamma$ growth. The control separates resonant amplification from the nonlinear source required to drive it at the beat.

D Efficiency of a memoryless transfer against a direct in-band drive

For the logistic family of memoryless transfers, compare balanced TI tones of polarization amplitude ε each with a direct in-band drive of amplitude 2ε, matching peak polarization. Their leading drive amplitudes are $|\sigma''|\varepsilon^2/2$ and $2\sigma'\varepsilon$, so $\eta_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$. This ratio concerns the specified transfer family and input matching, not TI in general or the dynamic detector of §2.3.

Writing $s = \sigma/(2e_0) \in (0,1)$ gives $\sigma' = 2e_0\rho s(1-s)$ and $\sigma'' = 2e_0\rho^2 s(1-s)(1-2s)$. Since $\mathrm{PEIX} = 4s(1-s)$, the leading ratio satisfies

$$\eta_{\mathrm{det}} \simeq \frac{\rho\,\varepsilon}{4}\sqrt{1-\mathrm{PEIX}} \;\le\; \frac{\rho\,\varepsilon}{4}. \tag{5}$$

It vanishes at the inflection, where the quadratic coefficient is zero, and approaches its bound in the saturated tails. This maximizes a ratio rather than an absolute response: both drives vanish in those tails. For the JR slope $\rho = 0.56\ \mathrm{mV}^{-1}$, the perturbative bound is 1.4% at $\varepsilon = 0.1$ mV and 0.28% at $\varepsilon = 0.02$ mV.

The slope is itself a parameter, so the question is what the whole sigmoid family can do. The logistic transfer depends on voltage only through $\rho(v - v_0)$, so writing

$$u = \rho\,\varepsilon, \qquad x = \rho\,(v^* - v_0), \tag{6}$$

we have $\sigma(v^* + \varepsilon q(t)) = 2e_0\,L(x + u\,q(t))$ for any normalized waveform q, with $L(z) = (1+e^{-z})^{-1}$. Every Fourier amplitude, and hence their ratio, is a function of (u, x) alone; the factor $2e_0$ cancels. Maximizing over v^* and ρ at fixed ε is therefore the same as maximizing over (u, x), and the family supremum is independent of field strength. Evaluating the exact cycle response rather than its quadratic truncation and optimizing x at each u, the ratio rises monotonically over the range explored and converges to the hard-threshold limit

$$\sup_{\rho,\,v^*} \eta_{\mathrm{det}} = 0.3479, \tag{7}$$

approached as $\rho \to \infty$ at a threshold offset δ (the distance from the operating point to the inflection) with $\delta/2\varepsilon = 0.523$. There is no finite optimal transfer width: narrower is always better, and reaching 95% of the ceiling requires $u \gtrsim 6$, that is $1/\rho \lesssim \varepsilon/6$.

Two features of that limit decide the reading. Such a detector is not weak. At the optimum, and at the Vieira-scale polarization $\varepsilon \simeq 0.1$ mV, its open-loop beat drive exceeds the physiological-slope sigmoid's by some three orders of magnitude for the same e_0 and waveform, because the field excursion is about twice the threshold offset and the transfer switches across its full range instead of responding linearly. The disparity grows as the field weakens, the smooth sigmoid's response falling as ε^2 while a threshold repositioned with the field keeps its excursion. The obstacle is the width. At Vieira et al.'s fields the ceiling is approached only for $1/\rho \lesssim 17\ \mu$V, and at human polarizations for $1/\rho \lesssim 3\ \mu$V, with the threshold offset held to the same order as the applied polarization. The width of a neural-mass sigmoid is set by the spread of thresholds and noise across $\sim 10^4$ neurons and is of order millivolts. A microvolt width lies far below that, and reads more naturally as an idealized microscopic threshold or rectifier than as a neural-mass sigmoid; we do not offer it as a measured single-cell property, real thresholds being themselves noisy and state-dependent. The regime in which the family approaches its ceiling is thus one in which the "population sigmoid" has ceased to describe a population.

The limit is a step detector, not an envelope follower. For a unit step at offset δ the carrier-averaged transduction is $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and zero below; with $\beta = \delta/2\varepsilon$ its tACS fundamental is $(2/\pi)\sqrt{1-\beta^2}$ and its TI coefficient reduces by parts to $\eta_{\mathrm{det}} = 2\beta\,[K(m) - E(m)]/(\pi\sqrt{1-\beta^2})$ with $m = 1-\beta^2$ and K, E the complete elliptic integrals, giving 0.3479 at $\beta = 0.523$ by direct quadrature and by the elliptic form alike (`threshold_bound_check.py`). At that optimum the

step's second-to-first harmonic magnitude ratio is 0.151, not the 1/5 of a linear envelope follower. A bounded nondecreasing transfer is a positive mixture of shifted steps and both Fourier coefficients are linear in the transfer, so the step optimum bounds the family; the maximum itself is a numerical evaluation, not a certified bound.

E Extension: the laminar column at low carriers

This extension drives the two-band laminar column (LaNMM, §2.1) with carriers of 25–80 Hz and polarizations up to 3 mV, outside the perturbative regime of the main text and below the carriers TI uses. It is included for what it shows about a band-limited intermediary between field and slow generator; it does not carry the kHz argument.

The single JR column isolates and lets us analyze the mechanism; the laminar model (LaNMM, §2.1) confirms it in a cortical column with its *own* two resonances. Driving one population with the two-tone field of Eq. (26) and reading the alpha-band power of the deep (P_1, alpha) and superficial (P_2, gamma) pyramidal potentials (Methods, Fig. S2), we recover the same envelope resonance, now carrying the column's intrinsic rhythms.

As a consistency check, driving the deep P_1 (alpha) loop directly reproduces the alpha-band beat resonance of the JR column on both the stable-focus and limit-cycle sides, with the sweep and its code in the repository; what the laminar model adds is the carrier-dependent structure below.

One qualification governs the laminar sweeps. They run to $\varepsilon = 3$ mV on P_2, above the sigmoid voltage scale $1/\rho \approx 1.8$ mV, so they sit deliberately in a nonlinear, non-perturbative regime. They are a demonstration of tongues and frequency capture, not a verification of the weak-field square law; that verification is the low-amplitude JR work (Fig. 6b), which fits the square law at 0.05–0.2 mV, one to two orders of magnitude below $1/\rho$.

Sweeping the field amplitude ε and beat frequency Δf at a fixed carrier produces *nonlinear Arnold tongues*: alpha power in P_1 is maximal when Δf matches the deep alpha ($\sim$10 Hz) and grows with ε (Figs. S1a,b and S7a,b), envelope resonance and phase locking of the slow generator to the recovered beat component, as in the JR resonance curves but now in a two-band column. Outside the tongue the drive *reduces* alpha-band power: with the field on P_1, 95% of the amplitude–beat grid and 97% of the carrier–beat grid lie below the field-free baseline (Fig. S7), the drive pulling the autonomous alpha out of the 8–12 Hz band or suppressing it; with the field on P_2 the fractions are 44% and 31% (Fig. S1). The earlier version of these maps showed only the positive part of the change.

The carrier dependence illustrates the two coupling limits of §2.2–§2.6 within this model's 25–80 Hz carrier range. Sweeping carrier f_c against beat Δf: when the field is delivered to the *superficial* loop (P_2), strong P_1 demodulation requires a *double* resonance—f_c near the P_2 gamma ($\approx$44 Hz) *and* Δf near the P_1 alpha ($\approx$10 Hz) (Fig. S1c,d): the fast loop acts as a tuned radio-frequency (RF) front end that must itself be excited before it can pass a demodulated envelope to P_1, and off that gamma resonance the response collapses to $\sim$7% of its peak. When the field is delivered *directly* to the slow generator (P_1), the picture inverts: the envelope-locked response persists at every carrier from 25 to 80 Hz, falling only to $\sim$25% of its peak, and the ridge sits at $\Delta f \approx 10.5$ Hz even for a 60 Hz carrier (Fig. S7c), broadband in f_c, as Eq. (10) predicts for direct coupling. This contrast requires the phase-sensitive read-out: because P_1 free-runs at its own alpha, band power alone buries a weak envelope-locked drive under an autonomous rhythm that is not phase-locked to the beat, and only the lock-in at Δf separates the two (Methods). TI's kHz carriers lie outside this model's range; what the contrast shows is that a carrier matched to a fast (gamma) network resonance couples best, while the envelope response persists, weaker, off resonance. This both confirms the JR mechanism and refines it: the "inert carrier" of §2.5 is the direct-coupling idealization, while a band-limited intermediary makes the carrier choice matter, consistent with frequency-tagging/intermodulation paradigms [7].

Concretely the LaNMM predicts that TI efficacy should (i) peak when Δf matches a region's intrinsic alpha; (ii) widen and strengthen with modulation amplitude (the tongue); and (iii) for fields delivered via superficial fast circuits, be largest when the carrier is matched to a gamma network resonance, for example two nearby gamma carriers summing to a $\sim$40 Hz-centered AM

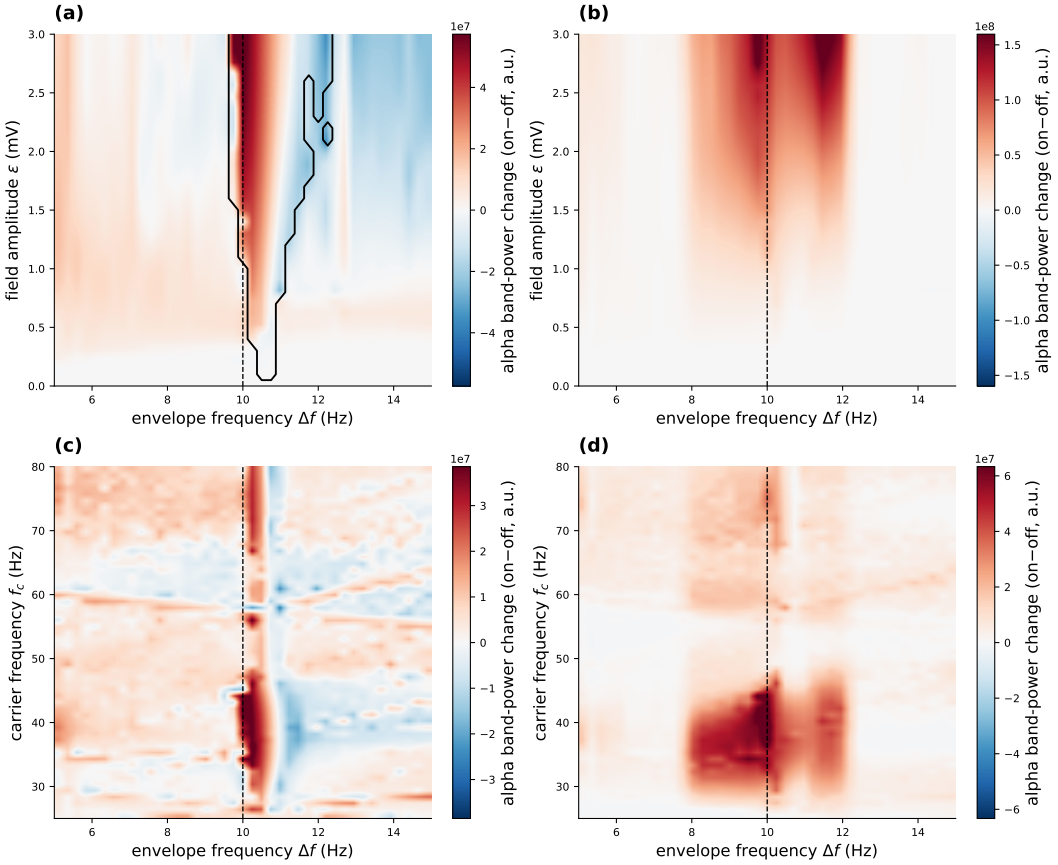


Figure S1: LaNMM Arnold tongues: two-tone TI field on the superficial P_2 (gamma) loop. The field enters the P_2 sigmoid argument as a membrane polarization ε (mV) with all external drives flat. Color shows the signed change of alpha-band (8–12 Hz) power from the unstimulated $\varepsilon = 0$ baseline, on a zero-centered scale (arbitrary units). *(a,b)* P_1 and P_2 enhancement vs. ε and Δf (carrier 40 Hz), peaking at the P_1 alpha (10.0 Hz, dashed). The black contour in *(a)* is the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), drawn because band power alone cannot separate a driven from an autonomous rhythm; it is a dominance screen, not a phase-capture test. The $\varepsilon = 0$ row is excluded, since with no drive the autonomous 10.67 Hz alpha already lies within the tolerance of two grid points (a resolution floor of 0.5 Hz); the set widens to 2.75 Hz at 3 mV. *(c,d)* Enhancement vs. carrier and Δf at $\varepsilon = 1.5$ mV: P_1 demodulation needs *both* a carrier near the P_2 gamma ($\approx$44 Hz) and Δf near the P_1 alpha—a double resonance.

field, while remaining present, weaker, for off-resonance carriers. The third is untested: existing TI studies of deep targets sweep the *beat* frequency rather than the carrier against a network resonance [8, 9], so they neither support nor contradict it.

F Vocabulary and metrics

The three terms (forced response, entrainment, gating) are defined in Methods; this appendix gives the metrics behind them. *(i) Forced response.* Below the Hopf the column does not oscillate on its own; the field elicits a driven response whose susceptibility we quantify by the lock-in amplitude of the read-out at the beat Δf (Eq. (25)), and there is no autonomous rhythm to entrain (J-curve, timing, tACS). *(ii) Frequency entrainment.* Above the Hopf the column free-runs at f_0; genuine entrainment means its dominant frequency is *pulled* to Δf. We screen for it with the dominant output frequency f_{out} (largest non-DC FFT peak) and the criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz, mapping the tongue and locking range (Fig. S4); f_{out} is used rather than a phase-locking value [10] because, when an autonomous rhythm and a driven component coexist, instantaneous-phase estimates conflate them. The dominance criterion is a screen, not a capture test: the largest peak can move to the drive frequency because a driven component outgrows a persisting autonomous one. A capture test would follow the unwrapped phase difference and require the absence of secular drift; the maps therefore label the set the frequency-dominance set. *(iii) Envelope locking / gating.* A periodic drive can instead modulate the *amplitude* of a persisting rhythm at the beat without capturing its frequency (phase–amplitude coupling) [11, 12]. We quantify this by the Δf-locked component of the population rate (lock-in of r_E at Δf, equivalently the rate *folded* on the Δf beat phase, averaged over many beat cycles as a function of phase within a single cycle, so that everything not locked to Δf averages away), which isolates the driven envelope from the intrinsic rhythm (the above-Hopf QIF panel, with its $\sim$11 Hz detuning, is a case of gating, not entrainment).

The Arnold-tongue protocol for Fig. S4 places the column on the limit-cycle side (input p between the SNIC and the Hopf, free-running at $f_0(p)$) and drives it with $s(t) = \varepsilon \cos \Omega t$; the implementation, including the grid widths and the clipping check, is in the released code. One subtlety: as onset is approached the cycle amplitude vanishes and the tongue merges into the forced-resonance bandwidth, so the supercritical locking range and the subcritical $1/\gamma$ susceptibility coincide there by construction—which is why responsiveness peaks at criticality on both sides.

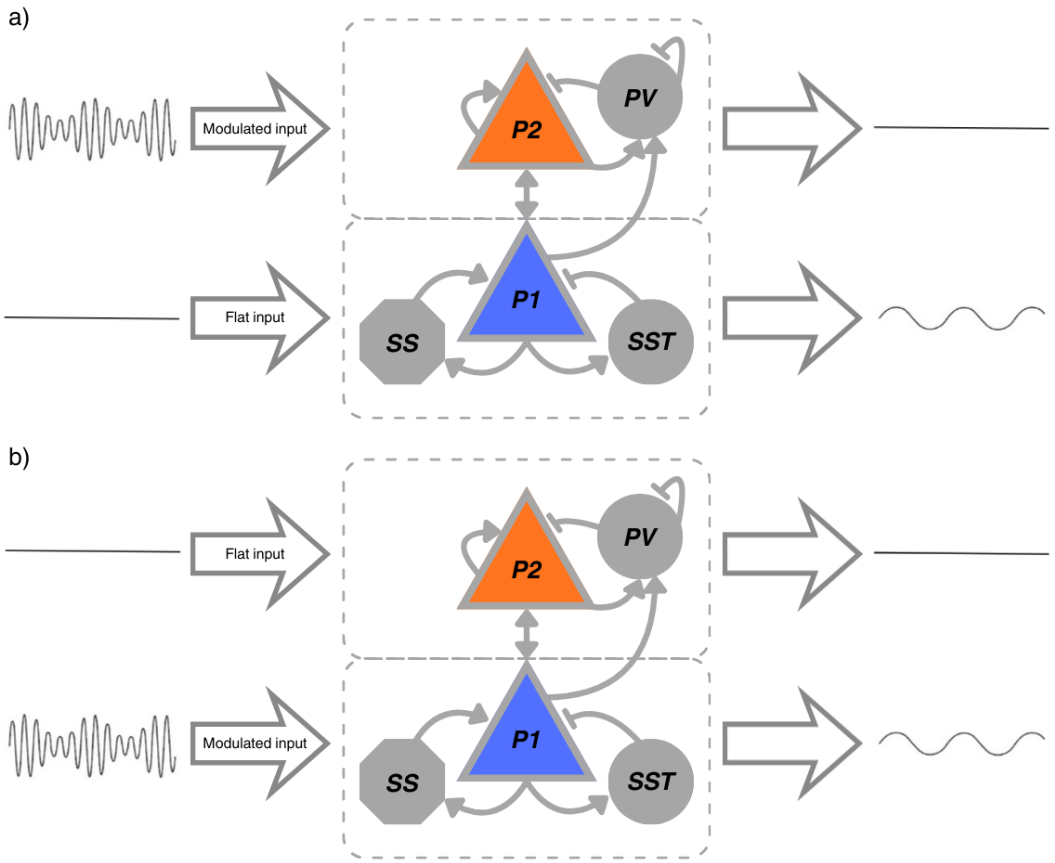


Figure S2: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the TI field drives P_2 while P_1 receives a flat drive; *(b)* the TI field drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the generated beat-frequency component is transferred to the deep loop: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency component (right traces).

G Supplementary figures

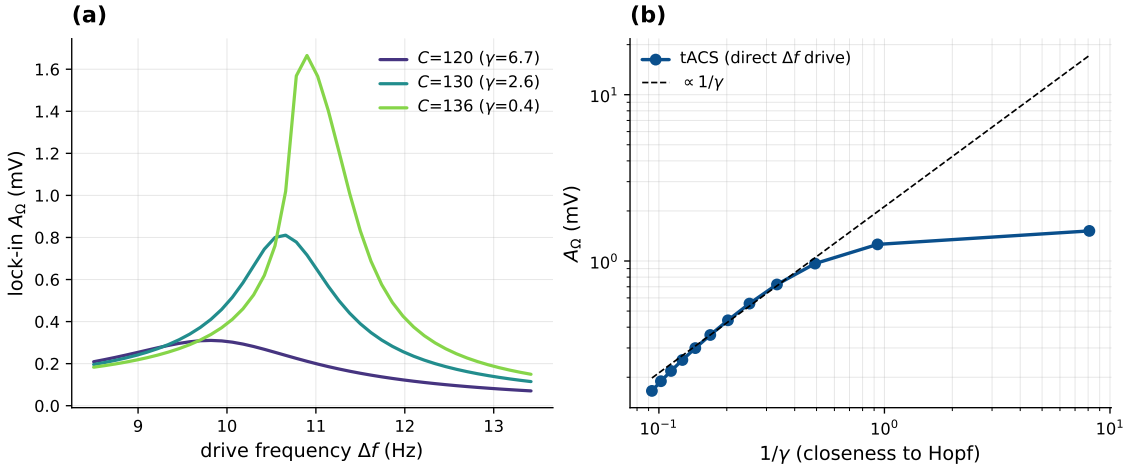


Figure S3: tACS shares the network amplifier (Jansen–Rit). A direct low-frequency (tACS) drive at Δf enters the column without a carrier, so it needs no demodulation, but feeds the same near-Hopf resonator. With the operating point pinned, sweeping the coupling $C\,(=J)$ toward the alpha Hopf: *(a)* the lock-in resonance at Δf sharpens and grows as $C \to C^*$; *(b)* the response grows as $1/\gamma$ (dashed) and saturates near criticality—the same network amplification as the demodulated TI drive (Fig. 9), but linear in the field rather than square-law. Throughout, the column is a stable focus ($\gamma > 0$).

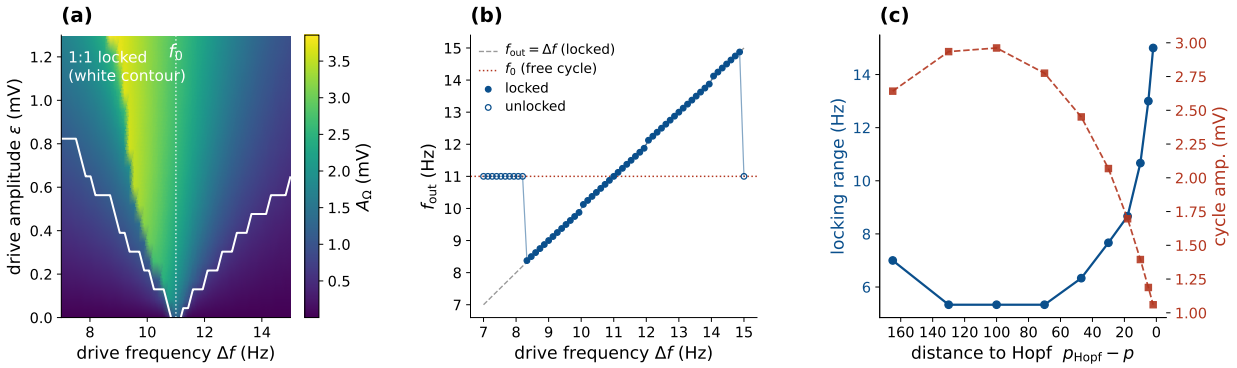


Figure S4: Above the Hopf: entrainment, not $1/\gamma$ amplification (Jansen–Rit). The column is set in its autonomous alpha limit cycle ($p = 250$, $f_0 \approx 11$ Hz) and driven by a direct in-band sinusoid. *(a)* The 1:1 Arnold tongue, where the dominant output frequency locks to Δf, is a wedge anchored at f_0 that widens with drive. *(b)* Inside the tongue the output follows Δf; outside, it falls back to f_0. *(c)* Toward the Hopf at fixed drive the locking range grows as the autonomous cycle amplitude vanishes (blue, left axis; red dashed, right axis). Above the Hopf amplification reads as tongue width rather than as the diverging $1/\gamma$ gain of the stable-focus side (Fig. 9): the column is most responsive near criticality on both sides.

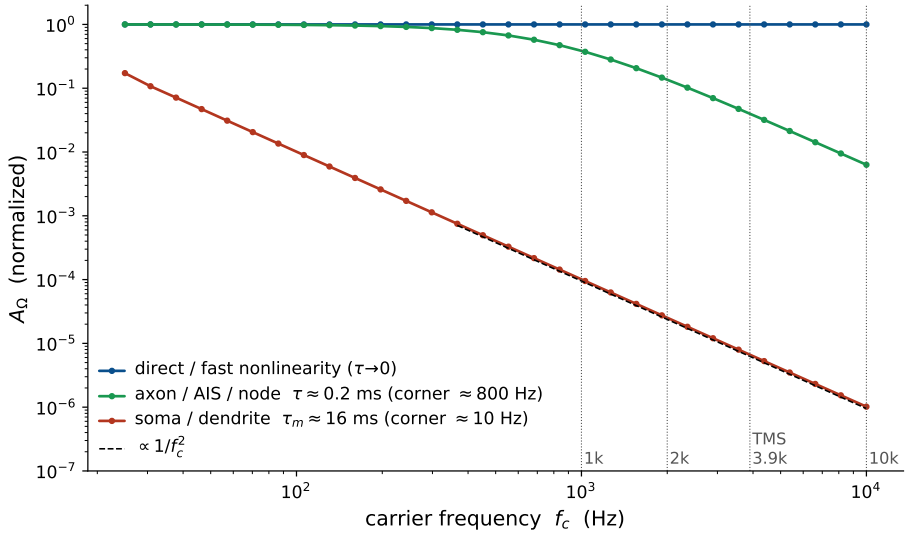


Figure S5: Carrier attenuation depends on the assumed access time (Jansen–Rit, open loop). Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide); the assumed fast response ($\tau = 0.2$ ms) retains about 0.14 of the direct response at 2 kHz and 0.04 at 3.9 kHz. These are sensitivity cases, not measurements of anatomical compartments.

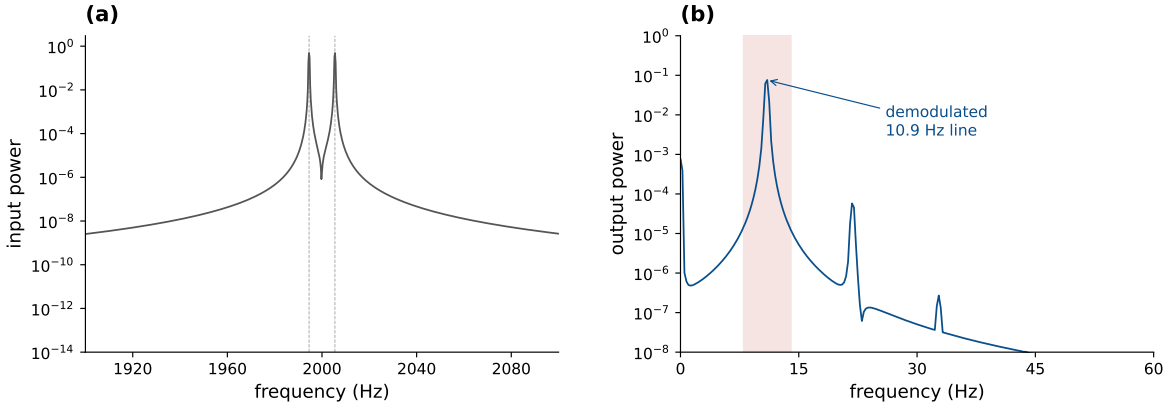


Figure S6: Two-tone TI at a kHz carrier (Jansen–Rit, assumed fast access). Two equal tones at $f_1 = 1994.54$ and $f_2 = 2005.46$ Hz, beating at approximately 10.9 Hz, pass through the assumed low-pass ($\tau = 0.2$ ms) into the recurrent column. *(a)* The input amplitude spectrum has lines only at f_1 and f_2; in the 8–14 Hz band it is 2.2×10^{-13}, zero to numerical precision. *(b)* The output carries a demodulated line at the beat, 10.9 Hz.

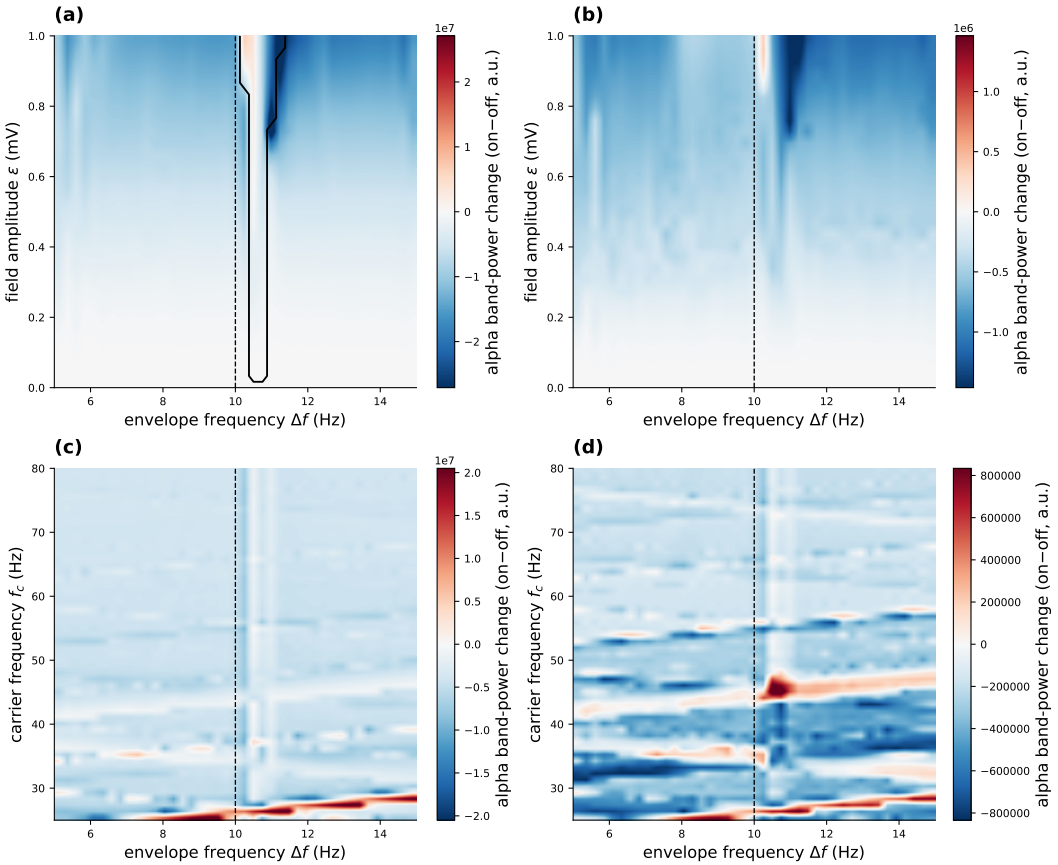


Figure S7: LaNMM Arnold tongues: two-tone TI field delivered directly to the deep P_1 (alpha) loop. As Fig. S1 but with the field on P_1. The deep loop is captured by a much weaker field than the superficial one, so the amplitude range is $\varepsilon \leq 1$ mV here against 3 mV there; above ≈2 mV the drive captures the whole Δf sweep and leaves the perturbative regime the theory describes. Color: signed change of alpha-band power from the $\varepsilon = 0$ baseline, zero-centered; most of the grid is suppression. *(a,b)* P_1 and P_2 enhancement vs. ε and Δf; the black contour in *(a)* outlines the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), computed from the dominant output frequency because band power alone cannot separate a driven from an autonomous rhythm; the $\varepsilon = 0$ row is excluded (resolution floor 0.5 Hz). The set widens to 1.25 Hz at $\varepsilon = 1$ mV. *(c,d)* Enhancement vs. carrier frequency and Δf at $\varepsilon = 0.5$ mV. P_1 responds across a *broad* carrier band—a vertical ridge at $\Delta f \approx 10$ Hz, the direct-coupling limit of Eq. (10). Read with the lock-in at Δf the response is retained at every carrier from 25 to 80 Hz (median 25% of peak), in contrast to the ∼7% collapse off resonance under superficial drive.

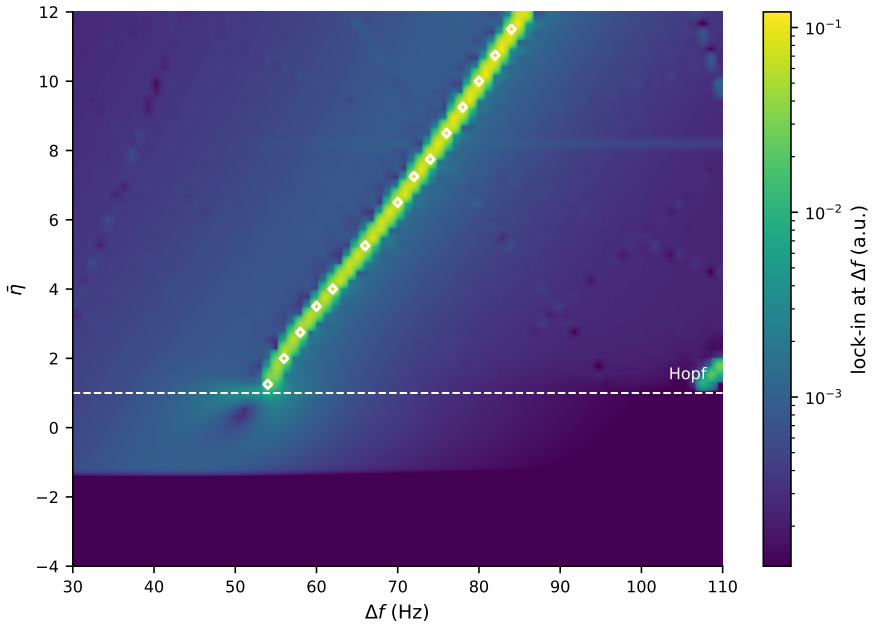


Figure S8: NMM2 PING resonance map (exact mean field). Demodulated response over the (Δf, background drive $\bar{\eta}$) plane, log color scale. The ridge follows the gamma natural frequency, with f_0 rising with $\bar{\eta}$ for these parameters. White markers outline the 1:1 captured set (Appendix F), drawn only above the Hopf (dashed line) where there is an autonomous rhythm to capture; below it the same color field is a forced response. Lock-in amplitude cannot separate the two, since both place power at Δf.

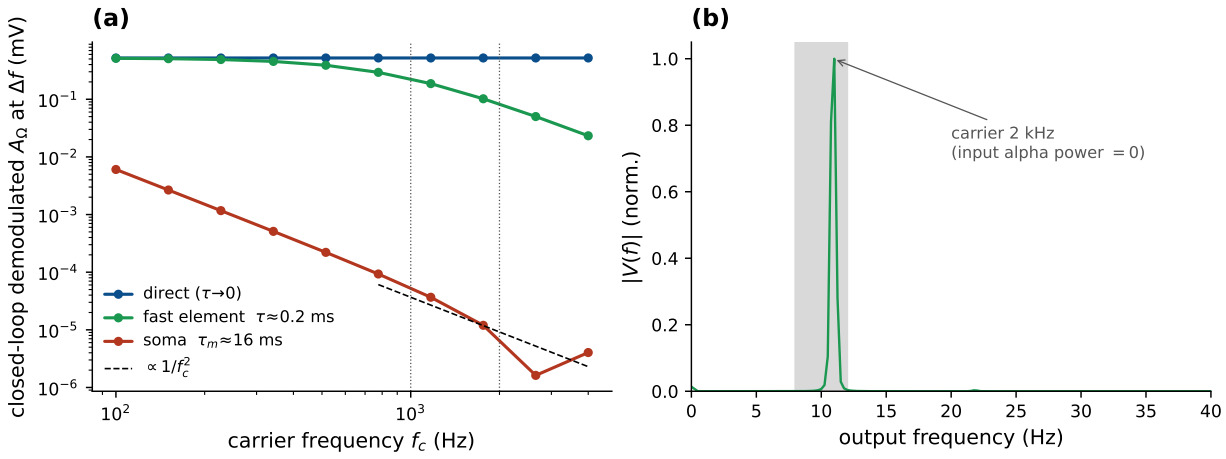


Figure S9: Demodulation at a genuine kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three access models: direct ($\tau \to 0$, carrier-independent), fast ($\tau = 0.2$ ms), and somatic ($\tau_m = 16$ ms; $1/f_c^2$ roll-off). The fast time is assumed rather than assigned to a measured compartment. *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: a clean alpha line at f_0, synthesized by the column although the input carries no alpha power.

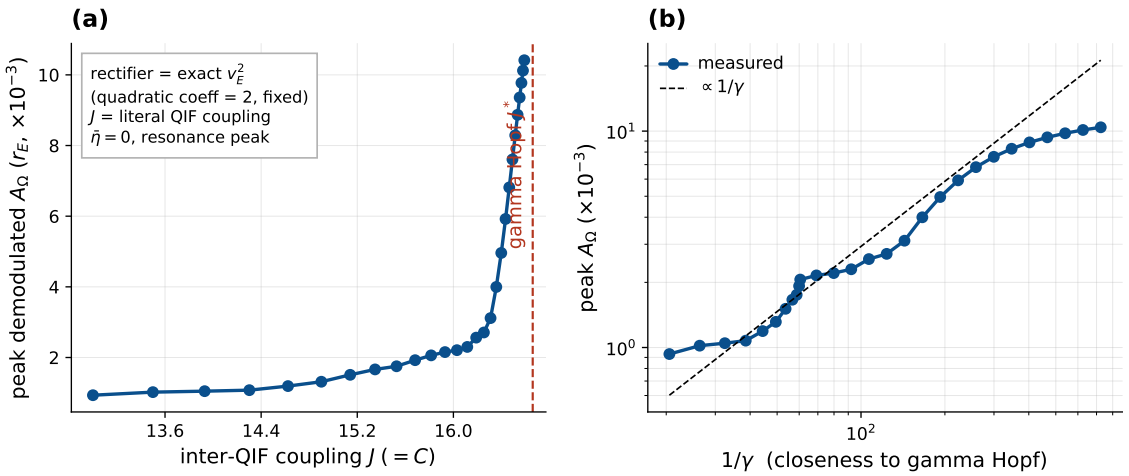


Figure S10: Coupling controls the amplification in the exact mean field (NMM2 PING, gamma). $J\,(=C)$ is the literal inter-QIF synaptic weight; at this carrier the generated source is operating-point independent (§2.3), so no pinning is needed. *(a)* The peak demodulated response of r_E grows as $J \to J^*$ (gamma Hopf, $\bar{\eta} = 0$; stable focus, $\gamma > 0$). *(b)* The gain grows as $1/\gamma$ (dashed), saturating near criticality—the JR law, now with coupling and rectifier both derived.

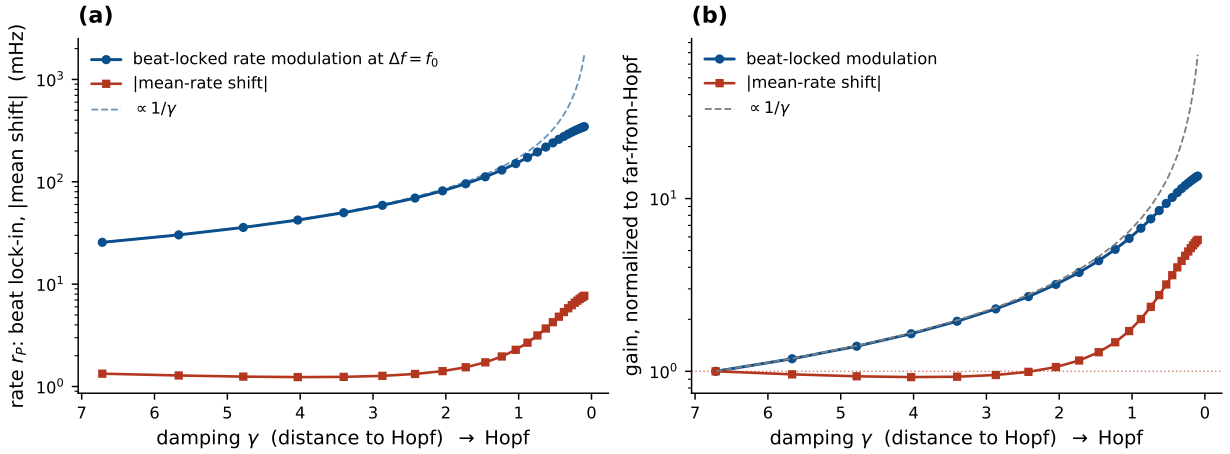


Figure S11: Envelope-locked modulation against mean-rate shift (Jansen–Rit, detection pinned). As the column is driven toward the Hopf (from the stable side, $\gamma > 0$) through the coupling at fixed $\sigma''(v^*)$, both quantities are read from the pyramidal rate $r_P = \sigma(v + s)$, in mHz. *(a)* The beat-locked modulation at $\Delta f = f_0$ (blue) is resonantly amplified and tracks $1/\gamma$ (dashed) before saturating; the magnitude of the mean-rate shift (red) stays at a few mHz and rises near the Hopf. *(b)* Both normalized to their far-from-Hopf values (log scale): the modulation gain follows $1/\gamma$; the mean-shift gain rises later and less. Modulation exceeds mean shift by 19× far from the Hopf and 45–66× near it.

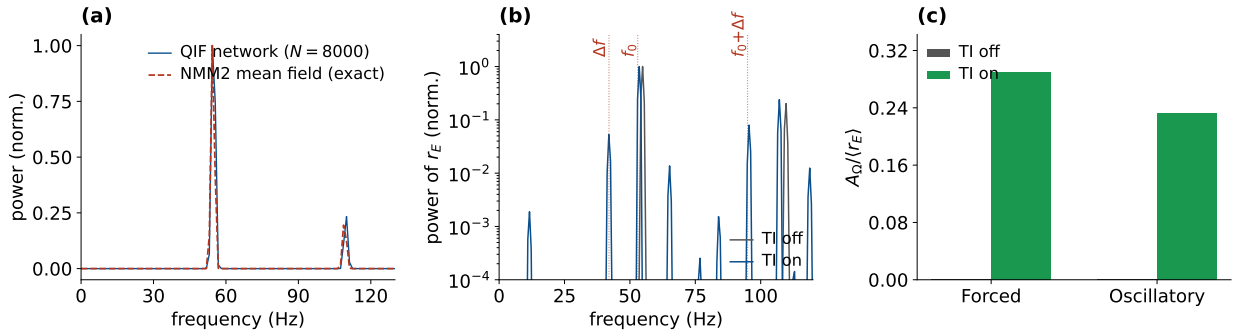


Figure S12: Timing, not rate, in the QIF network. *(a)* Field-free power spectrum of $r_E(t)$, normalized: the spiking network ($N = 8000$) matches the exact NMM2 mean field in gamma frequency (∼53 Hz); a normalized spectrum does not test the mean rate. *(b)* Intermodulation above the Hopf (TI off vs. on, log power): the nonlinearity mixes the imposed $\Delta f = 42$ Hz beat with the intrinsic 53 Hz gamma, raising the direct Δf line and the sum sideband near 95 Hz, both absent with TI off. This is the sideband signature of gating rather than entrainment. *(c)* The Δf modulation depth $A_\Omega/\langle r_E \rangle$, off vs. on: below 0.003 rising to 0.29 (forced) and 0.23 (oscillatory), while the mean rate rises by 8–10%. Modulation depth is reported rather than the off→on lock-in ratio, whose denominator is the field-free floor.

References

[1] Martin Vinck, Marijn van Wingerden, Thilo Womelsdorf, Pascal Fries, and Cyriel M. A. Pennartz. The pairwise phase consistency: a bias-free measure of rhythmic neuronal synchronization. *NeuroImage*, 51(1):112–122, 2010.

[2] Francesca Castaldo, Raul de Palma Aristides, Pau Clusella, Jordi Garcia-Ojalvo, and Giulio Ruffini. Rosetta Stone of neural mass models. *Physics Reports*, 1189:1–49, 2026. arXiv:2512.10982.

[3] Giulio Ruffini, Edmundo Lopez-Sola, Raul Palma, Roser Sanchez-Todo, Jakub Vohryzek, Francesca Castaldo, and Karl Friston. Cross-frequency coupling as a neural substrate for prediction error evaluation: a laminar neural mass modeling approach. *bioRxiv*, 2025. Preprint 2025.03.19.644090.

[4] Roser Sanchez-Todo, André M. Bastos, Edmundo Lopez-Sola, Borja Mercadal, Emiliano Santarnecchi, Earl K. Miller, Gustavo Deco, and Giulio Ruffini. A physical neural mass model framework for the analysis of oscillatory generators from laminar electrophysiological recordings. *NeuroImage*, 270:119938, 2023.

[5] Ben H. Jansen and Vincent G. Rit. Electroencephalogram and visual evoked potential generation in a mathematical model of coupled cortical columns. *Biological Cybernetics*, 73(4):357–366, 1995.

[6] Arkady Pikovsky, Michael Rosenblum, and Jürgen Kurths. *Synchronization: A Universal Concept in Nonlinear Sciences*. Cambridge University Press, 2001.

[7] Nathan Gordon, Jakob Hohwy, Matthew J. Davidson, Jeroen J. A. van Boxtel, and Naotsugu Tsuchiya. From intermodulation components to visual perception and cognition—a review. *NeuroImage*, 199:480–494, 2019.

[8] Zeinab Esmaeilpour, Greg Kronberg, Davide Reato, Lucas C. Parra, and Marom Bikson. Temporal interference stimulation targets deep brain regions by modulating neural oscillations. *Brain Stimulation*, 14(1):55–65, 2021.

[9] Ines R. Violante, Ketevan Alania, Antonino M. Cassarà, Esra Neufeld, Emma Acerbo, Romain Carron, Adam Williamson, Danielle L. Kurtin, Edward Rhodes, Adam Hampshire, Niels Kuster, Edward S. Boyden, Alvaro Pascual-Leone, and Nir Grossman. Non-invasive temporal interference electrical stimulation of the human hippocampus. *Nature Neuroscience*, 26(11):1994–2004, 2023.

[10] Jean-Philippe Lachaux, Eugenio Rodriguez, Jacques Martinerie, and Francisco J. Varela. Measuring phase synchrony in brain signals. *Human Brain Mapping*, 8(4):194–208, 1999.

[11] Ryan T. Canolty and Robert T. Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010.

[12] Adriano B. L. Tort, Robert Komorowski, Howard Eichenbaum, and Nancy Kopell. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2):1195–1210, 2010.